

E62109375_206

The equation $x^4 + x - y^3 + 8y - 1 = 0$ has no integer solutions

Size $h = 43$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + x - y^3 + 8y - 1 = 0 \tag{1}$$

has no integer solutions.

The proof uses 43 auxiliary polynomials $H_0, \dots, H_{42}$, 6 of which are constants, and a graph on these vertices with 113 edges; to each edge $\{i, j\}$ we attach the Hilbert symbol $(H_i, H_j)_v$ of the values at a hypothetical solution. By the product formula, the product of these symbols over all edges and all places is 1. We show that the product over the edges is 1 at every odd prime outside a finite set S , compute it at the primes of S and at the real place, and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only basic properties of the Legendre symbol and of the Hilbert symbol.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. *Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.*

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. *Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form*

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. *Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.*

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. *Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then*

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. *Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.*

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 The graph

Put $F(x, y) = x^4 + x - y^3 + 8y - 1$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$H_0 = x^2 - x + y,$$

$$H_1 = y,$$

$$H_2 = -x + y,$$

$$H_3 = y - 1,$$

$$H_4 = y + 1,$$

$$H_5 = x + y,$$

$$H_6 = -2x + y,$$

$$H_7 = -x + y - 1,$$

$$H_8 = -x + y + 1,$$

$$H_9 = y + 2,$$

$$H_{10} = x + y - 1,$$

$$H_{11} = x + y + 1,$$

$$H_{12} = -2x + y - 1,$$

$$H_{13} = -2x + y + 1,$$

$$H_{14} = -x + y - 2,$$

$$H_{15} = -x + y + 2,$$

$$H_{16} = x + y - 2,$$

$$H_{17} = x + y + 2,$$

$$H_{18} = 2x + y - 1,$$

$$H_{19} = 2x + y + 1,$$

$$H_{20} = -2x + y + 2,$$

$$H_{21} = x^2 - y,$$

$$H_{22} = x^2 + y,$$

$$H_{23} = x^2 - y + 1,$$

$$H_{24} = x^2 + y - 1,$$

$$H_{25} = x^2 + x + y,$$

$$H_{26} = x^2 - x - y - 1,$$

$$H_{27} = x^2 + x - y + 1,$$

$$H_{28} = x^2 - x + y + 1,$$

$$H_{29} = x^2 + y - 2,$$

$$H_{30} = x^2 + x + y - 1,$$

$$H_{31} = x^2 + x + y + 1,$$

$$H_{32} = x^2 + 2x + y,$$

$$H_{33} = x^2 - 2x - y - 1,$$

$$H_{34} = x^2 + x - y - 2,$$

$$H_{35} = x^2 + 2x - y - 1,$$

$$H_{36} = x^2 + 2x - y + 1,$$

$$H_{37} = 1,$$

$$\begin{aligned}
H_{38} &= 2, \\
H_{39} &= 3, \\
H_{40} &= -1, \\
H_{41} &= -3, \\
H_{42} &= -2.
\end{aligned}$$

The edges are $\{0, 2\}, \{0, 3\}, \{0, 4\}, \{0, 5\}, \{0, 6\}, \{0, 9\}, \{0, 15\}, \{0, 26\}, \{1, 13\}, \{1, 18\}, \{1, 25\}, \{1, 27\}, \{1, 29\}, \{2, 15\}, \{3, 17\}, \{3, 25\}, \{3, 29\}, \{4, 19\}, \{4, 23\}, \{4, 31\}, \{5, 18\}, \{5, 23\}, \{5, 32\}, \{5, 33\}, \{6, 9\}, \{6, 15\}, \{6, 22\}, \{7, 10\}, \{7, 14\}, \{7, 21\}, \{7, 24\}, \{7, 25\}, \{7, 26\}, \{7, 29\}, \{7, 34\}, \{7, 35\}, \{8, 18\}, \{8, 20\}, \{8, 21\}, \{8, 29\}, \{8, 34\}, \{8, 36\}, \{9, 14\}, \{9, 19\}, \{9, 20\}, \{9, 23\}, \{9, 27\}, \{10, 11\}, \{10, 16\}, \{10, 21\}, \{10, 22\}, \{10, 25\}, \{10, 26\}, \{10, 27\}, \{10, 34\}, \{11, 22\}, \{11, 27\}, \{11, 36\}, \{12, 21\}, \{12, 27\}, \{12, 30\}, \{12, 31\}, \{13, 27\}, \{13, 28\}, \{14, 26\}, \{14, 35\}, \{15, 23\}, \{15, 24\}, \{15, 28\}, \{15, 30\}, \{15, 35\}, \{15, 36\}, \{16, 36\}, \{17, 18\}, \{17, 22\}, \{17, 25\}, \{18, 24\}, \{18, 26\}, \{18, 27\}, \{19, 25\}, \{19, 32\}, \{20, 34\}, \{21, 32\}, \{21, 33\}, \{22, 31\}, \{23, 35\}, \{23, 36\}, \{24, 33\}, \{24, 35\}, \{25, 34\}, \{25, 36\}, \{26, 28\}, \{26, 30\}, \{27, 29\}, \{27, 30\}, \{27, 35\}, \{29, 32\}, \{29, 35\}, \{31, 32\}, \{33, 36\}, \{2, 38\}, \{3, 39\}, \{4, 40\}, \{5, 41\}, \{6, 38\}, \{15, 39\}, \{16, 42\}, \{20, 38\}, \{21, 38\}, \{26, 40\}, \{28, 38\}, \{29, 38\}, \{31, 38\}. For a vertex i , we write $j \sim i$ if $\{i, j\}$ is an edge. For every non-constant vertex i , Table 1 gives a positive integer m_i and polynomials W_i , A_i and B_i with integer coefficients such that$

$$m_i^2 \prod_{j \sim i} H_j - W_i^2 = A_i F + B_i H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij} F + \beta_{ij} H_i + \gamma_{ij} H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient -1 , and its discriminant is

$$\Delta(x) = -27x^8 - 54x^5 + 54x^4 - 27x^2 + 54x + 2021.$$

For $i \in I = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. The non-constant H_i are non-zero by Step 2.

Step 2: signs. *The signs of H_i for $i \in I$.* If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $-(z - y)$ for $z \neq y$. For $i \in I$ and $z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $\varphi_i(x)$. Expanding $\Delta(10 + t)$ and the $\varphi_i(10 + t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-10 - t)$ and the $\varphi_i(-10 - t)$. Hence, for $x \geq 10$ and for $x \leq -10$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-9 \leq x \leq 9$. Checking

the residues of y , we find that for $x \in \{-9, -6, -5, -4, -1, 0, 1, 4, 5, 6, 9\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , for $x \in \{-8, -7, -2, 2, 3, 7\}$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , and for $x = -3$ and $x = 8$, we have $F(x, y) \not\equiv 0 \pmod{19}$ for every y . Hence (1) has no solution with $-9 \leq x \leq 9$.

For a place v , put $B_v = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{37, 38, 39\}$ and $H_i < 0$ for $i \in \{40, 41, 42\}$. The other signs are given in Table 3 for $x \geq 10$ and for $x \leq -10$. In both cases, counting the edges whose two values are negative gives $B_\infty = 1$.

Step 3: odd primes outside S . Let $S = \{2, 3, 5, 7\}$ be the set of primes dividing 2, the m_i , the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p = 1$. For an edge $\{i, j\}$ such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim i} H_j$ is not divisible by p . By (2), $m_i^2 \Pi_i \equiv W_i^2 \pmod{p}$, and $p \nmid m_i$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p = 1$.

Step 4: the primes in S . Let $p \in S$. By Lemma 4, $B_p = (-1)^N$, where $N = \sum_{\{i,j\}} \beta_p(c_p(H_i), c_p(H_j))$, the sum over the edges. Suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Then $H_i \equiv H_i(a, b) \pmod{p^k}$, and Lemma 5 gives a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$; for a constant vertex, we take $b_i = c_p(H_i)$ and $D_i = 0$. If the hypotheses of Lemma 6 hold for the edges, then $N = \sum_{\{i,j\}} \beta_p(b_i, b_j)$, so B_p is determined by a, b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until these hypotheses hold, or until no solution remains. Table 4 lists the resulting terminal classes together with the value of B_p ; every integer solution lies in one of them. The result is $B_2 = 1, B_3 = -1, B_5 = 1, B_7 = 1$.

Step 5: contradiction. By Theorem 2(iv), applied to each edge, $\prod_v B_v = 1$, the product over all primes and $v = \infty$. By Step 3, the factors with $p \notin S$ are 1, so $\prod_{p \in S} B_p = B_\infty = 1$ by Step 2. But by Step 4, $\prod_{p \in S} B_p = -1$. This contradiction proves the theorem. $\square$

References

- [1] K. Ireland and M. Rosen, *A Classical Introduction to Modern Number Theory*, 2nd ed., Graduate Texts in Mathematics 84, Springer, New York, 1990.
- [2] J.-P. Serre, *A Course in Arithmetic*, Graduate Texts in Mathematics 7, Springer, New York, 1973.

Table 1: The data of the identities (2).

$i = 0, m = 1$, neighbours 2, 3, 4, 5, 6, 9, 15, 26	
W	$5x^5 + 3x^4 - 29x^3 + 32x - 4$
A	$-25x^6 - 30x^5 + 281x^4 + 9x^3y^3 + x^3y^2 + 10x^3y + 191x^3 - 27x^2y^3$ $-4x^2y^2 + 202x^2y - 1152x^2 + 3xy^4 - 18xy^3 + 9xy^2 + 228xy + 674x + 2y^5$ $-9y^4 - 18y^3 + 26y^2 - 786y + 16$
B	$-9x^5y^3 - x^5y^2 - 10x^5y + 8x^5 - 9x^4y^3 - x^4y^2 - 10x^4y + 8x^4$ $+9x^3y^4 - 8x^3y^3 + 9x^3y^2 - 18x^3y - 1141x^3 - x^2y^5 + 26x^2y^4$ $+274x^2y^3 - 11x^2y^2 - 1490x^2y + 817x^2 + 6xy^6 - 23xy^5 - 46xy^4$ $+484xy^3 - 55xy^2 - 2055xy + 1936x + y^7 - 14y^6 - 41y^5 + 101y^4 - 643y^3$ $-206y^2 + 6314y - 914$
$i = 1, m = 1$, neighbours 13, 18, 25, 27, 29	
W	$2x^3 + 2x^2 - x - 2$
A	$-4x^4 - 4x^3 + x^2y^2 - 4x^2y + 3x^2 + 2xy^2 - 4xy + 6x - 3y^3 + 4y^2 + 27y + 4$
B	$-2x^3y^2 - 8x^3y + 42x^3 + x^2y^4 - 5x^2y^3 + 2x^2y^2 + 16x^2y - 15x^2$ $+2xy^4 - 4xy^3 - 10xy^2 + 40xy - 88x - 3y^5 + 3y^4 + 54y^3 - 32y^2 - 215y - 3$
$i = 2, m = 1$, neighbours 38, 0, 15	
W	$2x$
A	0
B	$2x^2 - 2x + 2y + 4$
$i = 3, m = 1$, neighbours 39, 0, 17, 25, 29	
W	$3x^3 - 3$
A	$3x^3 + 3x^2y - 3x^2 + 9xy - 9x + 9y^2 + 9y - 21$
B	$3x^3y^2 + 12x^3y - 30x^3 + 3x^2y^3 + 9x^2y^2 - 12x^2y - 18x^2 + 9xy^3$ $+3xy^2 - 84xy - 12x + 9y^4 + 21y^3 - 72y^2 - 147y + 30$
$i = 4, m = 1$, neighbours 40, 0, 19, 23, 31	
W	$-2x^2 + 2x$
A	$-2x^3 - x^2y - x^2 - 2xy - 2x - y^2 - 2y - 1$
B	$-2x^3y^2 + 4x^3y + 8x^3 - x^2y^3 + x^2y^2 + 5x^2y - x^2 - 2xy^3 + 2xy^2$ $+14xy - y^4 + 8y^2 + 6y - 1$
$i = 5, m = 1$, neighbours 41, 0, 18, 23, 32, 33	
W	$-3x^3 - 6x^2 + 18x + 3$
A	$-6x^5 - 3x^4y + 9x^4 + 3x^3y + 21x^3 + 30x^2y - 27x^2 - 6xy^3 + 21xy^2 + 12xy$ $-27x - 3y^4 + 15y^3 + 6y^2 - 72y + 9$
B	$-6x^3y^2 + 63x^3y + 24x^3 + 3x^2y^4 + 24x^2y^3 - 60x^2y^2 - 228x^2y$ $+279x^2 - 3xy^5 + 6xy^4 + 21xy^3 - 9xy^2 - 57xy - 282x - 3y^6 + 15y^5 + 27y^4$ $-192y^3 - 21y^2 + 579y - 144$
$i = 6, m = 1$, neighbours 38, 0, 9, 15, 22	
W	$-2x^3 - 6x^2 - 4x$
A	$-4x^2 - 2xy - 28x + 2y^2 + 14y$
B	$2x^3y + 20x^3 + 4x^2y^2 + 18x^2y + 26x^2 + 2xy^3 - 14xy - 4x + 2y^4 + 16y^3$ $-8y^2 - 102y + 14$
$i = 7, m = 1$, neighbours 10, 14, 21, 24, 25, 26, 29, 34, 35	
W	$8x^3 + 64x^2 + 120x + 60$
A	$-x^{12} - 4x^{11} + x^{10}y^2 - 2x^{10}y + 5x^{10} + 3x^9y^2 - 6x^9y + 34x^9$ $-2x^8y^3 + 23x^8y - 6x^7y^3 - 6x^7y^2 + 80x^7y - 116x^7 + x^6y^5$ $-5x^6y^4 + 6x^6y^3 + 5x^6y^2 - 79x^6y - 50x^6 + 3x^5y^5 - 13x^5y^4$ $+36x^5y^3 + 21x^5y^2 - 415x^5y + 207x^5 - 2x^4y^6 + 3x^4y^5$ $+40x^4y^4 - 24x^4y^3 - 171x^4y^2 + 20x^4y + 160x^4 - 6x^3y^6$ $-3x^3y^5 + 139x^3y^4 - 160x^3y^3 - 510x^3y^2 + 1185x^3y - 211x^3$ $+x^2y^8 - 5x^2y^7 + x^2y^6 + 31x^2y^5 - 116x^2y^4 - 69x^2y^3$ $+635x^2y^2 + 639x^2y - 353x^2 + 3xy^8 - 13xy^7 + 19xy^6 + 103xy^5 - 737xy^4$ $+223xy^3 + 3282xy^2 - 2085xy - 919x - y^9 - 4y^8 + 17y^7 + 155y^6 - 57y^5$ $-1585y^4 - 468y^3 + 3248y^2 - 1151y - 754$
B	$x^3y^9 - 7x^3y^8 - 36x^3y^7 + 197x^3y^6 + 409x^3y^5 - 1698x^3y^4$ $-2054x^3y^3 + 3069x^3y^2 + 1228x^3y + 4883x^3 + x^2y^{10} - 2x^2y^9$ $-25x^2y^8 + 43x^2y^7 + 352x^2y^6 - 469x^2y^5 - 3038x^2y^4$ $+1413x^2y^3 + 8639x^2y^2 + 1459x^2y + 11303x^2 + 2xy^{10} - 16xy^9 + 420xy^7$ $-487xy^6 - 4036xy^5 + 2645xy^4 + 12880xy^3 + 609xy^2 + 4312xy + 10211x - y^{11}$ $-5y^{10} + 20y^9 + 207y^8 + 8y^7 - 2806y^6 - 2651y^5 + 13229y^4 + 14219y^3$ $-12991y^2 - 527y + 4354$

$i = 8, m = 1$, neighbours 18, 20, 21, 29, 34, 36

$W \quad 5x^3 - 13x^2 - 29x + 34$

$A \quad -4x^6 - 6x^5 + x^4y^2 + 9x^4y + 20x^4 + 3x^3y^2 + 3x^3y + 28x^3$
 $\quad - 6x^2y^3 - 3x^2y^2 - 11x^2y - 59x^2 - 9xy^3 - xy^2 + 30xy + 74x + y^5 + 7y^4$
 $\quad + 13y^3 - 5y^2 - 220y - 62$

$B \quad -2x^3y^4 + 2x^3y^3 + 56x^3y^2 - 120x^3y - 203x^3 - 5x^2y^5$
 $\quad + 39x^2y^3 - 40x^2y^2 - 119x^2y + 792x^2 + xy^6 - 4xy^5 + 3xy^4 + 28xy^3$
 $\quad - 229xy^2 + 258xy + 890x + y^7 + 6y^6 - 2y^5 - 58y^4 - 252y^3 + 241y^2 + 1502y - 1218$

$i = 9, m = 1$, neighbours 0, 6, 14, 19, 20, 23, 27

$W \quad 6x^3 + 30x + 18$

$A \quad -8x^6 + 12x^5y - 12x^5 - 2x^4y^2 + 16x^4y + 4x^4 - 3x^3y^3 - 15x^3y^2$
 $\quad + 10x^3y + 12x^3 + x^2y^4 - 5x^2y^3 + 18x^2y^2 - 8x^2 + 15xy^4 - 28xy^3$
 $\quad - 41xy^2 + 66xy - 12x - 3y^5 + 12y^4 - 17y^3 - 36y^2 - 60y - 364$

$B \quad -3x^3y^5 - 4x^3y^4 + 58x^3y^3 - 54x^3y^2 + 23x^3y - 104x^3 + x^2y^6$
 $\quad - 8x^2y^5 + 30x^2y^4 - 20x^2y^3 - 105x^2y^2 + 242x^2y - 452x^2 + 15xy^6$
 $\quad - 63xy^5 - 28xy^4 + 364xy^3 - 427xy^2 + 299xy - 364x - 3y^7 + 19y^6 - 32y^5 - 76y^4$
 $\quad + 245y^3 - 579y^2 + 1598y - 344$

$i = 10, m = 1$, neighbours 7, 11, 16, 21, 22, 25, 26, 27, 34

$W \quad -4x^3 - 20x^2 + 36x$

$A \quad -x^{11} - x^{10}y - 2x^{10} + x^9y^2 - 2x^9y + 5x^9 + x^8y^3 + 2x^8y^2 + x^8y$
 $\quad + 15x^8 - x^7y^3 + 3x^7y^2 + 11x^7y + 3x^7 - 3x^6y^4 - 3x^6y^3$
 $\quad + 15x^6y^2 + 15x^6y - 31x^6 + x^5y^5 - 7x^5y^4 - 4x^5y^3 + 14x^5y^2$
 $\quad - 41x^5y - 39x^5 + x^4y^6 + x^4y^5 - 9x^4y^4 - 8x^4y^3 - 17x^4y^2$
 $\quad - 127x^4y + 8x^4 - x^3y^6 + 7x^3y^5 + 21x^3y^4 - 27x^3y^3 - 98x^3y^2$
 $\quad - 30x^3y + 53x^3 - 3x^2y^7 + x^2y^6 + 41x^2y^5 + 31x^2y^4 - 144x^2y^3$
 $\quad - 107x^2y^2 + 311x^2y + 8x^2 + xy^8 - 7xy^7 - 10xy^6 + 72xy^5 - 5xy^4$
 $\quad - 161xy^3 + 379xy^2 + 410xy - 203x + y^9 - y^8 - 22y^7 + 17y^6 + 131y^5 - 166y^4$
 $\quad - 89y^3 + 714y^2 - 625y + 40$

$B \quad 2x^3y^8 + 4x^3y^7 - 31x^3y^6 - 68x^3y^5 + 67x^3y^4 + 240x^3y^3$
 $\quad + 387x^3y^2 + 468x^3y - 197x^3 - 3x^2y^9 + 5x^2y^8 + 63x^2y^7$
 $\quad - 49x^2y^6 - 429x^2y^5 - x^2y^4 + 809x^2y^3 + 149x^2y^2 - 100x^2y$
 $\quad + 1288x^2 - 7xy^9 + 4xy^8 + 138xy^7 - 75xy^6 - 818xy^5 + 701xy^4 + 1534xy^3$
 $\quad - 2741xy^2 - 1551xy + 203x + y^{11} - 30y^9 - 3y^8 + 303y^7 - 27y^6 - 1147y^5$
 $\quad + 1035y^4 + 956y^3 - 4809y^2 + 905y - 40$

$i = 11, m = 1$, neighbours 10, 22, 27, 36

$W \quad -x^3 + x^2 - x - 4$

$A \quad x^3 + x^2y + x^2 + 2xy + 3x - y^2 + 2y + 4$

$B \quad 3x^3y - 9x^3 + x^2y^3 - 4x^2y^2 + 2x^2y + x^2 + 3xy^3 - 6xy^2 - 10xy + 3x$
 $\quad - y^4 + 4y^3 + 5y^2 - 19y - 12$

$i = 12, m = 1$, neighbours 21, 27, 30, 31

$W \quad 9x^3 + 12x^2 - 11x - 6$

$A \quad x^4 + 3x^3 - 77x^2 - xy - 214x + y^3 + 20y^2 + 59y - 86$

$B \quad 11x^3y^2 + 34x^3y - 69x^3 + 4x^2y^3 + 13x^2y^2 - 39x^2y - 191x^2 + 2xy^4$
 $\quad + 43xy^3 - 25xy^2 - 385xy + 16x + y^5 + 21y^4 + 73y^3 - 173y^2 - 626y + 122$

$i = 13, m = 1$, neighbours 1, 27, 28

$W \quad 2x^3 - 7x^2 - 5x + 4$

$A \quad -4x^2 + 28x - 15y - 13$

$B \quad -8x^3y + 8x^3 - 4x^2y^2 + 45x^2 - 2xy^2 + 6xy + 23x - 15y^3 + y^2 + 119y - 29$

$i = 14, m = 1$, neighbours 7, 9, 26, 35

$W \quad -x^3 + 2x^2 + 4x + 1$

$A \quad -x^2 - xy + 2x + y^2 + 1$

$B \quad x^3 - 3x^2y^2 - 8x^2y + 5x^2 + 3xy^2 + 9xy + 3x + y^4 + 3y^3 + 2y^2 + 6y + 1$

$i = 15, m = 1$, neighbours 39, 0, 2, 6, 23, 24, 28, 30, 35, 36

$W \quad 174x^3 + 324x^2 - 444x - 522$

$A \quad 6x^{12} - 9x^{11}y + 18x^{11} + 3x^{10}y^2 - 21x^{10}y - 6x^{10} + 57x^9y - 36x^9$
 $\quad + 9x^8y^3 - 93x^8y^2 + 72x^8y + 6x^8 - 9x^7y^4 + 69x^7y^3 - 51x^7y^2$
 $\quad - 195x^7y + 18x^7 + 3x^6y^5 - 30x^6y^4 - 6x^6y^3 + 354x^6y^2 - 75x^6y$
 $\quad - 6x^6 + 81x^5y^4 - 339x^5y^3 - 165x^5y^2 + 315x^5y - 6x^5 + 9x^4y^6$
 $\quad - 102x^4y^5 + 231x^4y^4 + 447x^4y^3 - 594x^4y^2 + 117x^4y - 18x^4$
 $\quad - 9x^3y^7 + 69x^3y^6 - 57x^3y^5 - 654x^3y^4 + 567x^3y^3$

$$\begin{aligned}
& +1113x^3y^2 - 171x^3y + 6x^3 + 3x^2y^8 - 30x^2y^7 - 21x^2y^6 \\
& +660x^2y^5 - 414x^2y^4 - 2091x^2y^3 + 858x^2y^2 - 357x^2y - 30234x^2 \\
& +81xy^7 - 396xy^6 - 420xy^5 + 2361xy^4 + 708xy^3 - 1845xy^2 + 99xy - 112746x \\
& +3y^9 - 54y^8 + 129y^7 + 630y^6 - 1896y^5 - 183y^4 + 8988y^3 - 41715y^2 + 23928y \\
& +250488 \\
B \quad & -6x^3y^9 + 48x^3y^8 - 48x^3y^7 - 432x^3y^6 + 366x^3y^5 \\
& +2217x^3y^4 + 4827x^3y^3 - 39189x^3y^2 + 23409x^3y + 200994x^3 \\
& +3x^2y^{10} - 33x^2y^9 + 36x^2y^8 + 624x^2y^7 - 1641x^2y^6 \\
& -2352x^2y^5 + 13878x^2y^4 - 23370x^2y^3 - 56418x^2y^2 + 247647x^2y \\
& -97626x^2 + 3xy^{10} + 15xy^9 - 204xy^8 + 30xy^7 + 1167xy^6 + 4017xy^5 \\
& -9063xy^4 - 63387xy^3 + 129249xy^2 + 156267xy - 418884x + 3y^{11} - 60y^{10} + 225y^9 \\
& +612y^8 - 4200y^7 + 3309y^6 + 18156y^5 - 78456y^4 + 108759y^3 + 375675y^2 \\
& -984489y - 10998 \\
i = 16, m = 2, \text{ neighbours } 42, 10, 36 \\
W \quad & x^3 + 2x^2 - 4x + 1 \\
A \quad & -x^2 - 4x - y + 3 \\
B \quad & x^3y + x^3 - x^2y^2 + x^2y + 9x^2 - 3xy^2 - 7xy - 7x - y^3 + y^2 + 18y - 5 \\
i = 17, m = 1, \text{ neighbours } 3, 18, 22, 25 \\
W \quad & -x^3 - 5x^2 - 5x + 3 \\
A \quad & -x^2 + 2xy - 12x + y^2 - 5y - 17 \\
B \quad & 5x^3y - 19x^3 + 3x^2y - 4x^2 + xy^3 - 5xy^2 + 4xy + 24x + y^4 - 6y^3 - 15y^2 + 72y \\
& -13 \\
i = 18, m = 1, \text{ neighbours } 1, 5, 8, 17, 24, 26, 27 \\
W \quad & -203x^3 + 230x^2 + 231x - 138 \\
A \quad & -x^5y - x^4y^2 - x^4y + x^3y^3 + 3x^3y^2 + 4x^3y + x^2y^4 + 4x^2y^3 \\
& +5x^2y^2 + 5x^2y - 41209x^2 - 2xy^4 - 3xy^3 + 4xy^2 - 2xy + 93380x + 25y^4 \\
& +501y^3 + 9677y^2 + 22799y - 36176 \\
B \quad & -x^3y^5 - 15x^3y^4 - 252x^3y^3 - 4840x^3y^2 - 11405x^3y + 38531x^3 \\
& +x^2y^6 + 8x^2y^5 + 114x^2y^4 + 2279x^2y^3 + 3265x^2y^2 - 24970x^2y \\
& -41274x^2 - 2xy^6 - 56xy^5 - 1095xy^4 - 21113xy^3 + 14098xy^2 + 172993xy - 82872x \\
& +26y^6 + 529y^5 + 10004y^4 + 28816y^3 - 84274y^2 - 256987y + 55220 \\
i = 19, m = 1, \text{ neighbours } 4, 9, 25, 32 \\
W \quad & -6x^3 - 20x^2 + 10x + 12 \\
A \quad & -36x^2 - 240x + y^2 - 123y - 106 \\
B \quad & 63x^3y - 86x^3 - 30x^2y^2 + 16x^2y + 336x^2 - 2xy^3 + 11xy^2 - 27xy + 126x + y^4 \\
& -123y^3 + 12y^2 + 975y - 250 \\
i = 20, m = 1, \text{ neighbours } 38, 8, 9, 34 \\
W \quad & -2x^2 + 2x \\
A \quad & 0 \\
B \quad & 2x^3 + 2x^2y + 2x - 2y^2 - 6y - 4 \\
i = 21, m = 1, \text{ neighbours } 38, 7, 8, 10, 12, 32, 33 \\
W \quad & -4x^5 - 2x^4 + 8x^3 + 8x^2 + 2x \\
A \quad & -16x^6 - 16x^5 + 56x^4 + 22x^3y^2 + 14x^3y - 128x^3 - 16x^2y^3 + 2x^2y^2 \\
& +120x^2y + 10x^2 - 22xy^3 + 10xy^2 + 112xy - 4x + 16y^4 + 14y^3 - 172y^2 - 26y \\
B \quad & -22x^5y^2 - 8x^5y + 242x^5 - 22x^3y^3 - 8x^3y^2 + 242x^3y - 192x^3 \\
& -14x^2y^4 + 38x^2y^3 + 120x^2y^2 - 408x^2y + 52x^2 + 22xy^5 - 8xy^4 \\
& -280xy^3 + 120xy^2 + 726xy - 172x - 16y^6 - 12y^5 + 298y^4 + 118y^3 - 1386y^2 - 34y \\
& +24 \\
i = 22, m = 1, \text{ neighbours } 6, 10, 11, 17, 31 \\
W \quad & x^5 + x^4 - x^3 - 3x^2 - 3x - 1 \\
A \quad & -x^6 - 2x^5 + x^4 - 2x^3y^2 - 2x^3y + 10x^3 - x^2y^3 + 8x^2y + 10x^2 \\
& -2xy^3 + 11xy + 3x - y^4 - y^3 + 6y^2 + 8y + 1 \\
B \quad & 2x^5y^2 + 2x^5y - x^5 - 2x^3y^3 - 2x^3y^2 + x^3y - 10x^3 + x^2y^4 \\
& +2x^2y^3 - 7x^2y^2 - 27x^2y - 27x^2 - 2xy^5 + 30xy^3 + 5xy^2 - 94xy - 18x \\
& -y^6 - y^5 + 15y^4 + 18y^3 - 47y^2 - 61y - 2 \\
i = 23, m = 1, \text{ neighbours } 4, 5, 9, 15, 35, 36 \\
W \quad & -2x^4 + 2x^3 - 10x^2 + 4x - 12 \\
A \quad & -4x^4 + 2x^3y + 12x^3 - x^2y^2 - 3x^2y - 46x^2 - 2xy^2 - 6xy + 16x + y^3 + 2y^2 \\
& +23y - 50 \\
B \quad & -2x^5y - 4x^5 - 2x^3y^2 - 2x^3y + 44x^3 + x^2y^4 + 2x^2y^3 + 16x^2y^2
\end{aligned}$$

$$\begin{aligned}
& +27x^2y - 122x^2 + 6xy^4 + 30xy^3 + 25xy^2 - 5xy + 158x - 2y^5 - 9y^4 - 31y^3 \\
& +35y^2 + 225y - 194 \\
i = 24, m = 1, \text{ neighbours } 7, 15, 18, 33, 35 \\
W & 2x^4 - 12x^2 + 2x + 12 \\
A & -4x^4 - x^3 - 3x^2y + 45x^2 - xy - 3x - 3y^2 + 24y - 45 \\
B & 3x^5 - 3x^3y - 13x^3 - 3x^2y^3 + 9x^2y^2 + 29x^2y - 130x^2 - xy^3 - xy^2 \\
& +13xy + 9x - 2y^4 + 24y^3 + y^2 - 198y + 187 \\
i = 25, m = 1, \text{ neighbours } 1, 3, 7, 10, 17, 19, 34, 36 \\
W & 6x^5 + 22x^4 + 4x^3 - 42x^2 - 32x - 4 \\
A & -36x^6 - 264x^5 - 2x^4y^2 + 2x^4y - 532x^4 + 3x^3y^4 + 14x^3y^3 \\
& -223x^3y^2 - 58x^3y + 1512x^3 + x^2y^4 - 49x^2y^3 - 5x^2y^2 + 305x^2y \\
& +2444x^2 + 3xy^5 + 2xy^4 - 304xy^3 + 238xy^2 + 2173xy + 640x - 6y^5 - 69y^4 \\
& +168y^3 + 445y^2 + 492y + 16 \\
B & -3x^5y^4 - 17x^5y^3 + 215x^5y^2 + 69x^5y - 1148x^5 + 3x^4y^4 \\
& +17x^4y^3 - 215x^4y^2 - 69x^4y + 1148x^4 + 3x^3y^5 + 17x^3y^4 \\
& -215x^3y^3 - 69x^3y^2 + 1148x^3y + 272x^3 + x^2y^6 + 5x^2y^5 \\
& +42x^2y^4 - 291x^2y^3 - 59x^2y^2 + 1492x^2y - 3664x^2 + 3xy^7 + 7xy^6 \\
& -264xy^5 + 17xy^4 + 3707xy^3 - 824xy^2 - 14244xy + 76x - 5y^7 - 68y^6 + 210y^5 \\
& +989y^4 - 908y^3 - 3379y^2 - 3499y + 368 \\
i = 26, m = 1, \text{ neighbours } 40, 0, 7, 10, 14, 18, 28, 30 \\
W & 2x^5 - 14x^4 + 6x^3 + 62x^2 - 32x - 48 \\
A & -6x^6 + x^5y + 55x^5 + 3x^4y^2 - 14x^4y - 211x^4 + 2x^3y^3 - 67x^3y^2 \\
& +98x^3y + 474x^3 - x^2y^4 + 9x^2y^3 - 46x^2y^2 + 91x^2y + 1755x^2 - xy^4 \\
& +63xy^3 - 13xy^2 - 449xy - 2333x + y^5 - 5y^4 + 75y^3 + 148y^2 - 2143y - 1922 \\
B & -3x^5y^3 + 72x^5y^2 - 104x^5y - 552x^5 - 3x^4y^3 + 72x^4y^2 - 104x^4y \\
& -552x^4 - 3x^3y^4 + 66x^3y^3 + 40x^3y^2 - 760x^3y + 65x^3 - x^2y^5 \\
& +12x^2y^4 - 249x^2y^3 + 116x^2y^2 + 2277x^2y - 4058x^2 + 2xy^6 - 72xy^5 \\
& +159xy^4 + 880xy^3 - 1465xy^2 - 2156xy - 741x - y^7 + 7y^6 - 79y^5 - 102y^4 \\
& +2848y^3 + 176y^2 - 17461y + 4226 \\
i = 27, m = 1, \text{ neighbours } 1, 9, 10, 11, 12, 13, 18, 29, 30, 35 \\
W & 22x^5 + 10x^4 - 36x^3 - 24x^2 + 12x + 36 \\
A & 8x^7y^2 + 16x^7y + 12x^6y^3 + 44x^6y^2 + 40x^6y - 484x^6 - 2x^5y^4 \\
& +36x^5y^3 + 42x^5y^2 - 76x^5y - 440x^5 - 7x^4y^5 - 5x^4y^4 + 9x^4y^3 \\
& -115x^4y^2 - 194x^4y + 1484x^4 - 25x^3y^5 - 50x^3y^4 + 191x^3y^3 \\
& +88x^3y^2 - 2332x^3y + 3208x^3 + x^2y^7 + 6x^2y^6 + 29x^2y^5 \\
& -49x^2y^4 - 884x^2y^3 - 249x^2y^2 + 4246x^2y - 1388x^2 + xy^7 + 15xy^6 \\
& +94xy^5 - 211xy^4 - 1045xy^3 + 498xy^2 + 5904xy - 3544x - y^8 + 2y^7 - 56y^6 \\
& +3y^5 + 1396y^4 - 867y^3 - 8275y^2 + 12242y - 4660 \\
B & 12x^5y^5 + 71x^5y^4 - 304x^5y^3 - 433x^5y^2 + 2470x^5y - 948x^5 \\
& -12x^4y^5 - 71x^4y^4 + 304x^4y^3 + 433x^4y^2 - 2470x^4y + 948x^4 \\
& +12x^3y^6 + 59x^3y^5 - 375x^3y^4 - 129x^3y^3 + 2903x^3y^2 - 3418x^3y \\
& -1932x^3 - 5x^2y^8 - x^2y^7 + 96x^2y^6 - 188x^2y^5 - 844x^2y^4 \\
& +3381x^2y^3 + 3795x^2y^2 - 15266x^2y + 3488x^2 - 16xy^8 - 50xy^7 + 337xy^6 \\
& +242xy^5 - 2837xy^4 + 3162xy^3 + 8842xy^2 - 15368xy + 6208x + y^{10} + 47y^8 + 54y^7 \\
& -1785y^6 - 823y^5 + 18602y^4 - 1985y^3 - 62645y^2 + 43570y - 5956 \\
i = 28, m = 1, \text{ neighbours } 38, 13, 15, 26 \\
W & 2x^3 - 2x \\
A & -4x^2 + 4x - 4y - 4 \\
B & -4x^3 + 4x^2y + 12x^2 + 2xy + 6x - 4y^3 - 2y^2 + 26y - 8 \\
i = 29, m = 1, \text{ neighbours } 38, 1, 3, 7, 8, 27, 32, 35 \\
W & -6x^5 - 18x^4 + 4x^3 + 46x^2 + 26x - 4 \\
A & -36x^6 - 216x^5 + 2x^4y^2 - 2x^4y - 276x^4 + 14x^3y^3 - 226x^3y^2 \\
& -790x^3y + 1230x^3 + 2x^2y^4 - 60x^2y^3 + 36x^2y^2 + 274x^2y + 2132x^2 \\
& +14xy^4 - 270xy^3 + 38xy^2 + 1730xy - 1668x + 2y^5 - 66y^4 + 58y^3 + 990y^2 - 224y \\
& -2976 \\
B & -18x^5y^3 + 240x^5y^2 + 780x^5y - 498x^5 + 18x^3y^4 - 276x^3y^3 \\
& -300x^3y^2 + 2058x^3y + 1252x^3 - 2x^2y^5 + 96x^2y^4 - 420x^2y^3 \\
& -734x^2y^2 + 3234x^2y - 998x^2 + 14xy^6 - 248xy^5 - 564xy^4 + 2842xy^3 \\
& +3380xy^2 - 8026xy - 758x + 2y^7 - 60y^6 - 80y^5 + 1356y^4 + 1962y^3 - 6912y^2 \\
& -11044y + 1496 \\
i = 30, m = 1, \text{ neighbours } 12, 15, 26, 27
\end{aligned}$$

W	$-2x^4 - 6x^3 + 2x^2 + 6x$
A	$-4x^4 - 11x^3 - 26x^2 - 3xy - 16x + 4y^2 - 27y + 23$
B	$-13x^5 + 13x^4 + 13x^3y + 39x^3 - 4x^2y^3 - 3x^2y^2 + 30x^2y + 22x^2$ $-7xy^3 - 4xy^2 + 41xy + 7x + 4y^4 - 22y^3 - 30y^2 + 187y - 25$
$i = 31, m = 2, \text{neighbours } 38, 4, 12, 22, 32$	
W	$-x^5 - x^4 - x^3 - x^2 + 4x - 4$
A	$-x^6 - 2x^5 - 3x^4 - x^3y^2 + 2x^3y - 9x^3 - x^2y^3 + 8x^2y + 6x^2 - 2xy^3$ $+xy^2 - 2xy - 2x - y^4 - y^3 + 5y^2 + 19y + 16$
B	$x^5y^2 - 2x^5y + 6x^5 - x^4y^2 + 2x^4y - 6x^4 - x^3y^3 + x^3y^2$ $-4x^3y - 15x^3 + x^2y^4 + 2x^2y^2 - 21x^2y - 30x^2 - xy^5 + 2xy^4 - 32xy^2$ $+74xy + 14x - y^6 + 13y^4 + 21y^3 - 46y^2 - 109y$
$i = 32, m = 2, \text{neighbours } 5, 19, 21, 29, 31$	
W	$4x^4 + 4x^3 - 4x^2 - 4x$
A	$-8x^4 + 8x^3y - 43x^3 + 4x^2y^2 + 24x^2y + 12x^2 + 16xy^2 + 5xy + 132x + 4y^3$ $+16y^2 + 66y$
B	$4x^5y + 23x^5 - 8x^4y - 46x^4 - 4x^3y^2 - 16x^3y + 12x^3 - 8x^2y^3$ $-8x^2y^2 + 40x^2y + 3x^2 + 8xy^4 - 35xy^3 - 56xy^2 + 220xy - 101x + 4y^5 + 12y^4$ $+34y^3 - 112y^2 - 504y + 66$
$i = 33, m = 1, \text{neighbours } 5, 21, 24, 36$	
W	$-4x^3 + 2x^2 + 6x + 2$
A	$-3x^3 + x^2y - 14x^2 + xy + 43x - y^2 + 19y + 20$
B	$4x^5 + 8x^4 + 4x^3y - 7x^3 - 3x^2y + 19x^2 - 3xy^3 + 3xy^2 + 14xy + 33x + y^4$ $-21y^3 - 5y^2 + 157y - 16$
$i = 34, m = 1, \text{neighbours } 7, 8, 10, 20, 25$	
W	$x^5 + x^4 - 5x^3 - x^2 + 7x - 3$
A	$-x^6 - 2x^5 + 9x^4 + x^3y^2 - 3x^3y + 14x^3 - x^2y^3 + 8x^2y - 38x^2$ $-2xy^3 - xy^2 + 22xy - 31x + y^4 + 4y^3 - 14y^2 - 12y + 49$
B	$-x^5y^2 + 3x^5y - x^5 + x^4y^2 - 3x^4y + x^4 - x^3y^3 + x^3y^2 + 5x^3y$ $+x^3 - x^2y^4 + 7x^2y^3 + 12x^2y^2 - 73x^2y + 26x^2 + xy^5 - 3xy^4$ $-4xy^3 + 26xy^2 - 37xy + 8x - y^6 - 2y^5 + 25y^4 - 8y^3 - 146y^2 + 211y - 20$
$i = 35, m = 16, \text{neighbours } 7, 14, 15, 23, 24, 27, 29$	
W	$-128x^5 - 192x^4 - 96x^3 + 224x^2 + 800x - 320$
A	$-256x^7 + 768x^6y - 16896x^6 - 768x^5y^2 + 1280x^5y - 48128x^5 + 256x^4y^3$ $-1024x^4y^2 - 1536x^4y - 58368x^4 + 768x^3y^4 - 1856x^3y^3$ $+28144x^3y^2 - 78356x^3y + 4343x^3 + 768x^2y^4 - 19456x^2y^3$ $+1536x^2y^2 + 131584x^2y + 310784x^2 - 768xy^5 + 1344xy^4 - 44464xy^3$ $-65436xy^2 + 562541xy + 371757x + 512y^5 + 12160y^4 + 71520y^3 - 194888y^2 - 472890y$ -209938
B	$-768x^5y^4 + 1856x^5y^3 - 26352x^5y^2 + 75028x^5y + 34057x^5$ $+1536x^4y^4 - 3712x^4y^3 + 52704x^4y^2 - 150056x^4y - 68114x^4$ $-768x^3y^5 - 256x^3y^4 - 21248x^3y^3 + 2560x^3y^2 + 240384x^3y$ $+71936x^3 + 256x^2y^6 + 1024x^2y^5 - 15616x^2y^4 - 29120x^2y^3$ $+67088x^2y^2 + 330004x^2y - 89079x^2 + 768xy^7 - 3136xy^6 + 18672xy^5$ $-54804xy^4 - 156745xy^3 + 751472xy^2 - 586228xy - 477211x - 512y^7 - 11904y^6$ $-53984y^5 + 343080y^4 + 687762y^3 - 2096416y^2 - 1507688y + 310290$
$i = 36, m = 8, \text{neighbours } 8, 11, 15, 16, 23, 25, 33$	
W	$-32x^5 - 176x^4 - 368x^3 - 336x^2 - 96x$
A	$-960x^6 - 11584x^5 - 128x^4y^2 - 192x^4y - 54208x^4 + 192x^3y^3$ $+9648x^3y^2 - 2020x^3y - 198549x^3 + 64x^2y^4 - 704x^2y^3 - 128x^2y^2$ $+7360x^2y - 249792x^2 - 64xy^4 - 12144xy^3 - 20588xy^2 + 97601xy - 69889x - 64y^5$ $+1280y^4 + 18720y^3 - 13544y^2 - 31586y + 25194$
B	$-192x^5y^3 - 9264x^5y^2 + 2788x^5y + 48917x^5 + 384x^4y^3$ $+18528x^4y^2 - 5576x^4y - 97834x^4 - 192x^3y^4 - 9408x^3y^3$ $-4160x^3y^2 + 51008x^3y - 21760x^3 - 128x^2y^5 - 832x^2y^4$ $-35776x^2y^3 - 11952x^2y^2 + 273764x^2y + 76629x^2 + 192xy^6 + 9584xy^5$ $-12196xy^4 - 198293xy^3 + 107376xy^2 + 959228xy - 145215x + 64y^7 - 1280y^6$ $-20640y^5 + 3464y^4 + 184170y^3 + 31968y^2 - 207688y + 25194$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 2\}, k_{ij} = 2$
α $-9x^4 - 18y - 2$
β $18x^6 - 9x^5y + 18x^5 - 27x^4y + 18x^4 - 18x^2y^2 + 99x^2y + 94x^2 - 18xy^2$
γ $-2xy + 94x + 18y^3 - 117y^2 - 96y + 162$
γ $9x^7 - 9x^4y^2 + 45x^4y - 27x^4 + 101x^3 + 68x - 36y^3 + 115y^2 + 240y - 164$
$\{i, j\} = \{0, 3\}, k_{ij} = 2 \cdot 3$
α 1
β $-x^2 - x + y - 1$
γ $2x + y^2 - 7$
$\{i, j\} = \{0, 4\}, k_{ij} = 2$
α x^3
β $-x^5 - x^4 + x^3y - x^3 - 4x^2 + 2x + 4y + 2$
γ $2x^4 + x^3y^2 - 2x^3y - 5x^3 + 2x - 4y + 2$
$\{i, j\} = \{0, 5\}, k_{ij} = 2^2 \cdot 3^2$
α $-5x^4 - 60y - 36$
β $10x^6 + 5x^5y + 10x^5 - 5x^4y + 10x^4 + 20x^2y^2 + 75x^2y - 118x^2 + 20xy^2$
γ $-74xy - 118x - 20y^3 - 55y^2 + 44y + 96$
γ $-5x^7 - 5x^4y^2 - 15x^4y + 15x^4 + 149x^3 - 214x - 40y^3 + 19y^2 + 436y + 132$
$\{i, j\} = \{0, 6\}, k_{ij} = 3^2$
α $8x^4 - 9$
β $-40x^6 + 16x^5y - 40x^5 + 56x^4y - 40x^4 - 8x^2y^2 - 48x^2y + 37x^2 - 8xy^2$
γ $-58xy + 37x + 8y^3 + 40y^2 - 95y + 153$
γ $-16x^7 + 8x^4y^2 - 48x^4y + 24x^4 + 10x^3 + 58x - 8y^3 - 49y^2 + 95y - 81$
$\{i, j\} = \{0, 9\}, k_{ij} = 3^4$
α $2x^3 - 7$
β $-2x^5 - 2x^4 + 2x^3y - 2x^3 - 5x^2 + x + 5y + 1$
γ $4x^4 + 2x^3y^2 - 6x^3y - 2x^3 + 4x - 7y^2 + 9y + 37$
$\{i, j\} = \{0, 15\}, k_{ij} = 2$
α $5x^4 - 10y - 54$
β $-10x^6 + 5x^5y + 15x^4y - 10x^2y^2 - 5x^2y + 58x^2 - 10xy^2 - 14xy + 40x + 10y^3$
γ $-5y^2 - 72y + 108$
γ $-5x^7 + 5x^4y^2 - 15x^4y + 5x^4 + 9x^3 + 68x - 20y^3 - 9y^2 + 170y - 26$
$\{i, j\} = \{0, 26\}, k_{ij} = 3^2$
α $-128x^3 - 8$
β $128x^5 - 64x^3y^2 - 32x^3y + 528x^3 + 392x^2 - 264x - 4y^2 - 194y - 135$
γ $128x^4 + 64x^3y^2 - 96x^3y - 400x^3 + 128x + 4y^2 - 198y - 1$
$\{i, j\} = \{1, 13\}, k_{ij} = 7$
α -16
β $8x^3 + 4x^2 + 2x - 16y^2 + 137$
γ $-8x^3 - 4x^2 - 2x - 9$
$\{i, j\} = \{1, 18\}, k_{ij} = 7$
α -16
β $-8x^3 - 4x^2 - 2x - 16y^2 + 119$
γ $8x^3 + 4x^2 + 2x + 9$
$\{i, j\} = \{1, 25\}, k_{ij} = 1$
α -1
β $-x^2 + x - y^2 + 7$
γ $x^2 - x + 1$
$\{i, j\} = \{1, 27\}, k_{ij} = 7$
α $-2x - 3$
β $2x^3 + x^2 - 2xy^2 + 13x - 3y^2 + 28$
γ $2x^3 + x^2 - 3x + 4$
$\{i, j\} = \{1, 29\}, k_{ij} = 2 \cdot 7$
α $-x^3 + 6$
β $-x^5 - x^3y^2 + 6x^3 + 5x^2 - 3x + 6y^2 - 38$
γ $x^5 + 2x^3 - 5x^2 + 3x - 10$
$\{i, j\} = \{2, 15\}, k_{ij} = 2$
α 0

β	-1
γ	1
$\{i, j\}$	$\{3, 17\}, k_{ij} = 2^2 \cdot 3 \cdot 7$
α	1
β	$x^3 - 3x^2 + 9x + y^2 + y - 33$
γ	$-x^3 + 3x^2 - 9x + 26$
$\{i, j\}$	$\{3, 25\}, k_{ij} = 2 \cdot 7$
α	$-x + 2$
β	$-x^3 + 3x^2 - xy^2 - xy + 5x + 2y^2 + 2y - 16$
γ	$x^3 - 3x^2 + 2x + 2$
$\{i, j\}$	$\{3, 29\}, k_{ij} = 2^4 \cdot 3$
α	$-x^3 + 7$
β	$-x^5 - x^3y^2 - x^3y + 6x^3 + 6x^2 - 7x + 7y^2 + 7y - 43$
γ	$x^5 + x^3 - 6x^2 + 7x - 6$
$\{i, j\}$	$\{4, 19\}, k_{ij} = 2^4$
α	-2
β	$-x^3 - 2y^2 + 2y + 13$
γ	$x^3 + 1$
$\{i, j\}$	$\{4, 23\}, k_{ij} = 2^2 \cdot 3^2$
α	$x^3 - 8$
β	$-x^5 + x^3y^2 - x^3y - 5x^3 + 7x^2 + 4x - 8y^2 + 8y + 42$
γ	$-x^5 + 2x^3 + 7x^2 + 4x - 14$
$\{i, j\}$	$\{4, 31\}, k_{ij} = 2^3$
α	-1
β	$-x^2 + x - y^2 + y + 6$
γ	$x^2 - x + 1$
$\{i, j\}$	$\{5, 18\}, k_{ij} = 2 \cdot 3$
α	-1
β	$x^3 - x^2y + xy^2 + y^3 - 4y^2 + 4y + 11$
γ	$-y^3 + 2y^2 - 2y - 5$
$\{i, j\}$	$\{5, 23\}, k_{ij} = 2 \cdot 3 \cdot 7$
α	$x^4 - 6y + 7$
β	$-x^7 + x^6y - x^5 + x^4y - x^4 - 7x^3 + 7x^2y + 6xy^2 + 6xy - 49x - 6y^3 - 6y^2$
γ	$+55y - 7$
γ	$-x^4y^2 + x^4 - 13y^2 - 6y + 49$
$\{i, j\}$	$\{5, 32\}, k_{ij} = 2^2 \cdot 3^2$
α	$-7x^4 + 49y - 36$
β	$7x^7 - 7x^6y - 14x^4y^2 + 7x^4 + 29x^3 - 29x^2y - 49xy^2 - 42xy + 301x + 49y^3$
γ	$-114y^2 - 434y + 638$
γ	$7x^4y^2 + 78y^2 + 42y - 301$
$\{i, j\}$	$\{5, 33\}, k_{ij} = 2$
α	$-3x^4 - 8y + 6$
β	$3x^7 - 5x^6y + 10x^5y - 15x^5 + 2x^4y^2 + 5x^4y + 33x^4 - 8x^3 - 10xy^2 + 78x$
γ	$-14y^3 - 12y^2 + 122y + 6$
γ	$2x^5y + 5x^4y^2 - 6x^4y + 15x^4 - 6x^2y^2 + 8x^2y + 14x^2 - 12xy^2 + 24xy$
γ	$+28x - 6y^3 - 12y^2 + 70y - 8$
$\{i, j\}$	$\{6, 9\}, k_{ij} = 2^4 \cdot 3^2$
α	-16
β	$-8x^3 - 4x^2y - 2xy^2 - y^3 - 8$
γ	$y^3 - 18y^2 + 36y + 64$
$\{i, j\}$	$\{6, 15\}, k_{ij} = 3^2 \cdot 5$
α	1
β	$y^3 + 3y^2 + 12y - 7$
γ	$x^3 + x^2y + 2x^2 + xy^2 + 4xy + 4x - y^3 - 12y + 23$
$\{i, j\}$	$\{6, 22\}, k_{ij} = 2^2 \cdot 3^3 \cdot 5^2$
α	$-92x^4 - 1058y - 2700$
β	$-46x^7 - 23x^6y - 230x^5y + 920x^5 - 115x^4y^2 + 460x^4y - 46x^4 - 1304x^3$
γ	$-652x^2y - 2116xy^2 + 1760xy + 10946x - 1058y^3 + 880y^2 + 4944y - 1350$
γ	$23x^4y^2 - 460x^4y + 1840x^4 - 3580y^2 + 3520y + 21892$

$$\begin{aligned}
&\{i, j\} = \{7, 10\}, k_{ij} = 2^2 \cdot 3 \\
&\alpha \quad 2 \\
&\beta \quad 2x^3 + 2x^2y - 2x^2 + 2xy^2 - 4xy + 2x + y^3 - 2y^2 + 4y - 7 \\
&\gamma \quad -y^3 + 4y^2 - 2y - 7 \\
&\{i, j\} = \{7, 14\}, k_{ij} = 1 \\
&\alpha \quad 0 \\
&\beta \quad 1 \\
&\gamma \quad -1 \\
&\{i, j\} = \{7, 21\}, k_{ij} = 2^2 \\
&\alpha \quad x^4 - 4y + 4 \\
&\beta \quad x^7 + x^6y - x^6 + 2x^5 + 2x^4y - x^4 + 4x^3 + 4x^2y - 4x^2 - 4xy^2 - 12xy + 12x \\
&\quad - 4y^3 - 8y^2 + 20y - 8 \\
&\gamma \quad -x^4y^2 + 2x^4y + x^4 - 8y^2 - 4y + 8 \\
&\{i, j\} = \{7, 24\}, k_{ij} = 2^4 \cdot 3 \\
&\alpha \quad -7x^4 + 49y - 41 \\
&\beta \quad -7x^7 - 7x^6y + 7x^6 - 14x^5y + 7x^5 - 14x^4y^2 + 21x^4y - 14x^4 - 34x^3 \\
&\quad - 34x^2y + 34x^2 + 49xy^2 - 26xy - 365x + 49y^3 - 75y^2 - 290y + 324 \\
&\gamma \quad 7x^4y^2 - 28x^4y + 14x^4 + 83y^2 - 94y - 331 \\
&\{i, j\} = \{7, 25\}, k_{ij} = 3^2 \\
&\alpha \quad 5x^4 - 25y - 14 \\
&\beta \quad 5x^7 + 5x^6y - 5x^6 + 15x^5y - 25x^5 + 10x^4y^2 - 15x^4y - 19x^3 - 19x^2y \\
&\quad + 19x^2 - 25xy^2 - 52xy + 320x - 25y^3 - 33y^2 + 257y + 5 \\
&\gamma \quad -5x^4y^2 + 25x^4y - 30x^4 - 6y^2 - 90y + 339 \\
&\{i, j\} = \{7, 26\}, k_{ij} = 2^3 \cdot 3^2 \\
&\alpha \quad -5x^4 + 50y + 94 \\
&\beta \quad -5x^7 - 5x^6y + 5x^6 + 5x^5y - 10x^4 + 104x^3 + 104x^2y - 104x^2 + 50xy^2 + 86xy \\
&\quad + 260x + 50y^3 + 90y^2 - 70y - 322 \\
&\gamma \quad 5x^4y^2 - 5x^4y + 5x^4 - 54y^2 + 294y + 156 \\
&\{i, j\} = \{7, 29\}, k_{ij} = 2 \cdot 7 \\
&\alpha \quad -2x^4 + 8y - 11 \\
&\beta \quad -2x^7 - 2x^6y + 2x^6 - 4x^5y + 4x^5 - 4x^4y^2 + 8x^4y - 6x^4 - 15x^3 \\
&\quad - 15x^2y + 15x^2 + 8xy^2 - 18xy - 22x + 8y^3 - 26y^2 + 4y + 11 \\
&\gamma \quad 2x^4y^2 - 8x^4y + 6x^4 + 23y^2 - 48y - 7 \\
&\{i, j\} = \{7, 34\}, k_{ij} = 3^2 \\
&\alpha \quad x^4 - 3y \\
&\beta \quad x^7 + x^6y - x^6 + x^5y - x^5 + x^4y + 3x^3 + 3x^2y - 3x^2 - 3xy^2 - 6xy - 3y^3 \\
&\quad - 9y^2 + 3y - 3 \\
&\gamma \quad -x^4y^2 + 3x^4y - 2x^4 - 6y^2 - 3 \\
&\{i, j\} = \{7, 35\}, k_{ij} = 2^2 \cdot 3^4 \\
&\alpha \quad x^4 - y + 37 \\
&\beta \quad x^7 + x^6y - x^6 + 2x^5y - 3x^5 + 3x^4y - 4x^4 + 44x^3 + 44x^2y - 44x^2 - xy^2 \\
&\quad + 80xy - 133x - y^3 - 9y^2 + 122y - 184 \\
&\gamma \quad -x^4y^2 + 4x^4y - 4x^4 - 45y^2 + 168y - 177 \\
&\{i, j\} = \{8, 18\}, k_{ij} = 2^5 \cdot 7 \\
&\alpha \quad -81 \\
&\beta \quad -81x^3 - 81x^2y - 81x^2 - 81xy^2 - 162xy - 81x - 27y^3 - 99y^2 + 33y + 448 \\
&\gamma \quad 27y^3 + 72y^2 + 138y + 305 \\
&\{i, j\} = \{8, 20\}, k_{ij} = 1 \\
&\alpha \quad 1 \\
&\beta \quad x^3 + x^2y + x^2 + xy^2 + 2xy + x - y^3 - 3y^2 - 9y - 24 \\
&\gamma \quad y^3 + 3y^2 + 6y + 13 \\
&\{i, j\} = \{8, 21\}, k_{ij} = 2 \cdot 3^3 \cdot 7 \\
&\alpha \quad 4x^4 - 32y - 41 \\
&\beta \quad 4x^7 + 4x^6y + 4x^6 - 8x^5 - 8x^4y - 4x^4 - 49x^3 - 49x^2y - 49x^2 - 32xy^2 \\
&\quad + 40xy + 378x - 32y^3 + 8y^2 + 386y + 337 \\
&\gamma \quad -4x^4y^2 - 8x^4y - 12x^4 + 17y^2 + 138y + 427 \\
&\{i, j\} = \{8, 29\}, k_{ij} = 2^3 \cdot 3 \\
&\alpha \quad x^4 + 8y - 1 \\
&\beta \quad x^7 + x^6y + x^6 + 2x^5y - 6x^5 + 2x^4y^2 - 4x^4y - 5x^4 + 7x^3 + 7x^2y
\end{aligned}$$

	$+7x^2 + 8xy^2 - 18xy - 10x + 8y^3 - 10y^2 - 20y - 11$
γ	$-x^4y^2 - 7x^4 + y^2 - 32y - 17$
$\{i, j\} = \{8, 34\}, k_{ij} = 3^2$	
α	$-x^4 + 5y + 10$
β	$-x^7 - x^6y - x^6 - x^5y + 3x^5 + 3x^4y + 6x^4 + 9x^3 + 9x^2y + 9x^2 + 5xy^2$
	$-54x + 5y^3 + y^2 - 67y - 107$
γ	$x^4y^2 + x^4y + 4x^4 - 4y^2 - 18y - 63$
$\{i, j\} = \{8, 36\}, k_{ij} = 2^5 \cdot 7$	
α	$x^4 - 7y - 21$
β	$x^7 + x^6y + x^6 + 2x^5y - x^5 + x^4y - 4x^4 - 28x^3 - 28x^2y - 28x^2 - 7xy^2$
	$-28xy + 35x - 7y^3 + 7y^2 + 56y + 140$
γ	$-x^4y^2 - 2x^4 + 21y^2 + 28y + 63$
$\{i, j\} = \{9, 14\}, k_{ij} = 3^5$	
α	1
β	$-x^3 + 4x^2 - 16x + y^2 - 2y + 59$
γ	$x^3 - 4x^2 + 16x - 63$
$\{i, j\} = \{9, 19\}, k_{ij} = 3^3 \cdot 5$	
α	-16
β	$-8x^3 - 4x^2 - 2x - 16y^2 + 32y + 55$
γ	$8x^3 + 4x^2 + 2x + 9$
$\{i, j\} = \{9, 20\}, k_{ij} = 2 \cdot 3^2$	
α	-2
β	$x^3 - 2y^2 + 4y + 9$
γ	$-x^3 - 1$
$\{i, j\} = \{9, 23\}, k_{ij} = 3^2$	
α	x^3
β	$-x^5 + x^3y^2 - 2x^3y - x^3 - x^2 + 3$
γ	$-x^5 + 3x^3 - x^2 + 3$
$\{i, j\} = \{9, 27\}, k_{ij} = 3^2 \cdot 5$	
α	$x^3 - 6$
β	$-x^5 + x^4 + x^3y^2 - 2x^3y - 2x^3 + 3x - 6y^2 + 12y + 21$
γ	$-x^5 + x^4 + 2x^3 + 3x - 3$
$\{i, j\} = \{10, 11\}, k_{ij} = 2$	
α	0
β	-1
γ	1
$\{i, j\} = \{10, 16\}, k_{ij} = 1$	
α	0
β	1
γ	-1
$\{i, j\} = \{10, 21\}, k_{ij} = 2 \cdot 5$	
α	$-x^4 + 2y + 3$
β	$x^7 - x^6y + x^6 + 2x^5 - 2x^4y + 3x^4 - x^3 + x^2y - x^2 - 2xy^2 - 8xy - 10x$
	$+2y^3 + 6y^2 - 13$
γ	$x^4y^2 - 2x^4y - x^4 + y^2 + 10y + 9$
$\{i, j\} = \{10, 22\}, k_{ij} = 2^3 \cdot 7$	
α	$x^4 - 8y + 8$
β	$-x^7 + x^6y - x^6 - 2x^5y + 2x^4y^2 - 2x^4y - x^4 - 8x^3 + 8x^2y - 8x^2$
	$+8xy^2 - 8xy - 56x - 8y^3 + 16y^2 + 56y - 64$
γ	$-x^4y^2 + 4x^4y - x^4 - 16y^2 + 24y + 48$
$\{i, j\} = \{10, 25\}, k_{ij} = 5^3$	
α	$x^4 - 5y + 14$
β	$-x^7 + x^6y - x^6 - x^5y - 3x^5 + 2x^4y^2 - x^4y - 6x^4 - 19x^3 + 19x^2y$
	$-19x^2 + 5xy^2 - 4xy - 72x - 5y^3 + 33y^2 + 31y - 139$
γ	$-x^4y^2 + 3x^4y + 2x^4 - 24y^2 + 42y + 53$
$\{i, j\} = \{10, 26\}, k_{ij} = 2^3$	
α	$x^4 - 2y + 2$
β	$-x^7 + x^6y - x^6 - x^5y + 2x^4y - 2x^4 - 4x^3 + 4x^2y - 4x^2 + 2xy^2 + 2xy$
	$-2y^3 - 6y^2 + 10y - 6$

$$\begin{aligned}
&\gamma \quad -x^4y^2 + 3x^4y - x^4 - 6y^2 + 6y - 4 \\
&\{i, j\} = \{10, 27\}, k_{ij} = 2^2 \cdot 3^2 \cdot 5 \\
&\alpha \quad -x^4 + 2y + 28 \\
&\beta \quad x^7 - x^6y + x^6 - x^5y + 2x^5 - 2x^4y + 4x^4 - 24x^3 + 24x^2y - 24x^2 - 2xy^2 \\
&\quad + 12xy - 76x + 2y^3 + 8y^2 + 50y - 156 \\
&\gamma \quad x^4y^2 - x^4y - x^4 - 22y^2 + 36y + 52 \\
&\{i, j\} = \{10, 34\}, k_{ij} = 3^4 \cdot 5 \\
&\alpha \quad -11x^4 + 121y + 416 \\
&\beta \quad 11x^7 - 11x^6y + 11x^6 - 11x^5y - 11x^5 + 11x^4y - 22x^4 - 405x^3 + 405x^2y \\
&\quad - 405x^2 - 121xy^2 - 90xy - 728x + 121y^3 + 253y^2 - 383y - 1467 \\
&\gamma \quad 11x^4y^2 - 11x^4y + 22x^4 - 284y^2 + 900y + 323 \\
&\{i, j\} = \{11, 22\}, k_{ij} = 2 \cdot 5 \\
&\alpha \quad -2x^4 - 8y - 5 \\
&\beta \quad 2x^7 - 2x^6y - 2x^6 + 4x^5y - 8x^5 - 4x^4y^2 + 4x^4y + 10x^4 - 7x^3 + 7x^2y \\
&\quad + 7x^2 + 8xy^2 - 26xy - 8y^3 + 18y^2 + 34y + 5 \\
&\gamma \quad 2x^4y^2 + 10x^4 - 15y^2 + 12y - 7 \\
&\{i, j\} = \{11, 27\}, k_{ij} = 2^2 \cdot 5^2 \\
&\alpha \quad x^4 - 8y - 2 \\
&\beta \quad -x^7 + x^6y + x^6 + x^5y + 4x^5 - 2x^4y + 8x^3 - 8x^2y - 8x^2 + 8xy^2 - 14xy - 96x \\
&\quad - 8y^3 + 6y^2 + 88y - 6 \\
&\gamma \quad -x^4y^2 - 3x^4y - 5x^4 + 30y + 104 \\
&\{i, j\} = \{11, 36\}, k_{ij} = 2^3 \cdot 5 \\
&\alpha \quad -7x^4 + 147y - 33 \\
&\beta \quad 7x^7 - 7x^6y - 7x^6 - 14x^5y - 63x^5 + 21x^4y - 70x^4 + 26x^3 - 26x^2y - 26x^2 \\
&\quad - 147xy^2 - 192xy + 655x + 147y^3 - 7y^2 - 1098y + 636 \\
&\gamma \quad 7x^4y^2 + 28x^4y + 70x^4 + 173y^2 + 244y - 629 \\
&\{i, j\} = \{12, 21\}, k_{ij} = 2^3 \\
&\alpha \quad 82x^4 - 410y - 396 \\
&\beta \quad 164x^7 + 82x^4y^2 - 82x^4y + 41x^4 - 1530x^3 - 1172x - 765y^2 - 26y + 388 \\
&\gamma \quad 246x^6 - 164x^5y + 164x^5 + 246x^4y + 410x^2y^2 - 369x^2y - 2541x^2 + 1530xy \\
&\quad - 1530x + 410y^3 - 369y^2 - 2541y - 2344 \\
&\{i, j\} = \{12, 27\}, k_{ij} = 2^3 \cdot 3^3 \\
&\alpha \quad -16x^4 - 64y + 100 \\
&\beta \quad -8x^7 - 4x^6y + 4x^6 - 28x^5y - 124x^5 - 12x^4y^2 - 80x^4y - 68x^4 - 30x^3 \\
&\quad - 15x^2y + 15x^2 - 128xy^2 - 297xy + 239x - 64y^3 - 205y^2 - 76y + 177 \\
&\gamma \quad 4x^4y^2 - 64x^4y - 244x^4 - 241y^2 - 624y + 493 \\
&\{i, j\} = \{12, 30\}, k_{ij} = 2^3 \cdot 3^5 \cdot 5^2 \\
&\alpha \quad 104x^4 + 2704y + 5396 \\
&\beta \quad 52x^7 + 26x^6y - 26x^6 + 286x^5y - 1378x^5 + 130x^4y^2 - 520x^4y - 650x^4 \\
&\quad + 4362x^3 + 2181x^2y - 2181x^2 + 5408xy^2 - 4297xy - 21369x + 2704y^3 - 535y^2 \\
&\quad - 9300y - 9077 \\
&\gamma \quad -26x^4y^2 + 624x^4y - 2782x^4 + 8635y^2 - 4232y - 44919 \\
&\{i, j\} = \{12, 31\}, k_{ij} = 2^6 \cdot 3 \\
&\alpha \quad -4x^4 - 48y - 68 \\
&\beta \quad -2x^7 - x^6y + x^6 - 11x^5y + 49x^5 - 5x^4y^2 + 18x^4y + 23x^4 + 6x^3 \\
&\quad + 3x^2y - 3x^2 - 96xy^2 + 257xy + 61x - 48y^3 + 79y^2 + 138y - 5 \\
&\gamma \quad x^4y^2 - 24x^4y + 99x^4 - 195y^2 + 520y + 119 \\
&\{i, j\} = \{13, 27\}, k_{ij} = 2 \cdot 7^2 \\
&\alpha \quad -y \\
&\beta \quad -2x^3y + 28x - y^3 - 4y^2 + 15y + 42 \\
&\gamma \quad -3x^2y + 2xy^2 + 5xy - 5y^2 - 2y + 56 \\
&\{i, j\} = \{13, 28\}, k_{ij} = 2^5 \cdot 3^2 \\
&\alpha \quad -20x^4 - 100y - 136 \\
&\beta \quad -10x^7 - 5x^6y - 5x^6 - 45x^5y + 165x^5 - 25x^4y^2 + 100x^4y - 95x^4 + 62x^3 \\
&\quad + 31x^2y + 31x^2 - 200xy^2 + 559xy + 157x - 100y^3 + 395y^2 - 220y - 131 \\
&\gamma \quad 5x^4y^2 - 80x^4y + 335x^4 - 431y^2 + 1056y + 283 \\
&\{i, j\} = \{14, 26\}, k_{ij} = 2 \cdot 3 \\
&\alpha \quad 2x^4 - 41y + 40 \\
&\beta \quad 2x^7 - 7x^6y + 14x^6 + 7x^5y - 21x^5 + 9x^4y^2 - 14x^4y + x^4 - 25x^3
\end{aligned}$$

	$+119x^2y - 238x^2 - 14xy^2 - 147xy + 322x - 32y^3 - 139y^2 + 345y + 234$
γ	$-9x^5y + 18x^5 + 7x^4y^2 - 30x^4y + 25x^4 + 9x^2y^2 - 9x^2y - 36x^2$ $+9xy^2 + 135xy - 324x + 9y^3 - 124y^2 + 419y - 514$
$\{i, j\} = \{14, 35\}, k_{ij} = 2^2 \cdot 3^2$	
α	$-2x^4 - 16y + 53$
β	$-2x^7 - 2x^6y + 4x^6 - 4x^5y + 6x^5 - 10x^4y + 14x^4 + 13x^3 + 13x^2y - 26x^2$ $-16xy^2 + 18xy + 65x - 16y^3 - 8y^2 + 153y - 51$
γ	$2x^4y^2 - 12x^4y + 14x^4 - 29y^2 + 70y + 13$
$\{i, j\} = \{15, 23\}, k_{ij} = 2 \cdot 3^2$	
α	-1
β	$-x^3 - x^2y - 2x^2 + 3x + 3y + 5$
γ	$y^2 + 4y + 7$
$\{i, j\} = \{15, 24\}, k_{ij} = 2^2 \cdot 3^2$	
α	$-35x^4 - 250y - 91$
β	$25x^7 - 5x^6y - 10x^6 - 10x^5y + 35x^5 - 40x^4y^2 - 15x^4y + 95x^4 - 395x^3$ $-203x^2y - 406x^2 - 100xy^2 + 14xy + 431x - 100y^3 + 6y^2 + 401y + 387$
γ	$60x^6 - 30x^5y - 60x^5 + 5x^4y^2 - 50x^4y + 115x^4 + 150x^2y^2 + 300x^2y$ $-414x^2 + 192xy + 384x - 150y^3 - 47y^2 + 1540y + 829$
$\{i, j\} = \{15, 28\}, k_{ij} = 2^2 \cdot 3$	
α	$-x^4 - 4y - 5$
β	$-x^7 - x^6y - 2x^6 - x^5y + 5x^5 - 2x^4y^2 + 5x^3 + 5x^2y + 10x^2 - 4xy^2$ $+11xy - x - 4y^3 + 12y^2 + 26y + 14$
γ	$x^4y^2 + 3x^4y + 9x^4 - 9y^2 - 9y - 21$
$\{i, j\} = \{15, 30\}, k_{ij} = 2 \cdot 3^4$	
α	$-7x^4 - 147y - 167$
β	$7x^7 - 7x^4y^2 - 28x^4y + 63x^4 - 176x^3 + 357x - 98y^3 + 29y^2 + 851y - 100$
γ	$14x^6 - 7x^5y - 28x^5 - 7x^4y + 42x^4 + 49x^2y^2 + 147x^2y - 100x^2 - 49xy^2$ $+29xy + 452x - 49y^3 - 49y^2 + 218y - 195$
$\{i, j\} = \{15, 35\}, k_{ij} = 2^3 \cdot 3^3$	
α	$x^4 + 8y + 7$
β	$x^7 + x^6y + 2x^6 + 2x^5y - 3x^5 - 3x^4y - 19x^4 - 25x^3 - 25x^2y - 50x^2$ $+8xy^2 - 58xy - 13x + 8y^3 + 24y^2 - 37y + 155$
γ	$-x^4y^2 - 2x^4y - 7x^4 + 33y^2 + 42y + 87$
$\{i, j\} = \{15, 36\}, k_{ij} = 2 \cdot 3^3 \cdot 5 \cdot 7$	
α	$x^4 + 2y - 79$
β	$x^7 + x^6y + 2x^6 + 2x^5y - x^5 - x^4y - 11x^4 - 107x^3 - 107x^2y - 214x^2$ $+2xy^2 - 196xy - 19x + 2y^3 + 26y^2 + 55y + 701$
γ	$-x^4y^2 - 2x^4y - 5x^4 + 109y^2 + 232y + 409$
$\{i, j\} = \{16, 36\}, k_{ij} = 2^3$	
α	1
β	$-x^3 + x^2y - 2x^2 + 2xy - 3x + 3y - 5$
γ	$-y^2 + 2y - 1$
$\{i, j\} = \{17, 18\}, k_{ij} = 2^3 \cdot 3 \cdot 7$	
α	1
β	$-x^3 + x^2y + 2x^2 - xy^2 - 4xy - 4x - y^3 + 2y^2 - 16y + 69$
γ	$y^3 + 2y^2 + 14y - 31$
$\{i, j\} = \{17, 22\}, k_{ij} = 2^4 \cdot 7^3$	
α	$-19x^4 - 361y - 1028$
β	$19x^7 - 19x^6y - 38x^6 + 38x^5y - 114x^5 - 38x^4y^2 + 38x^4y + 247x^4$ $+515x^3 - 515x^2y - 1030x^2 + 361xy^2 - 262xy - 601x - 361y^3 - 460y^2 + 1486y + 2230$
γ	$19x^4y^2 + 38x^4y + 190x^4 + 154y^2 + 2322y + 2661$
$\{i, j\} = \{17, 25\}, k_{ij} = 7^2$	
α	$-x^4 - 7y - 17$
β	$x^7 - x^6y - 2x^6 + x^5y - 6x^5 - 2x^4y^2 + x^4y + 3x^4 + 10x^3 - 10x^2y$ $-20x^2 + 7xy^2 - 11xy - 39x - 7y^3 - 6y^2 + 17y + 16$
γ	$x^4y^2 + 3x^4y + 10x^4 + 3y^2 + 51y + 79$
$\{i, j\} = \{18, 24\}, k_{ij} = 2^5 \cdot 3 \cdot 5$	
α	$-28x^4 - 98y + 178$
β	$14x^7 - 7x^6y + 7x^6 + 70x^5y - 182x^5 - 35x^4y^2 + 126x^4y - 77x^4 + 44x^3$

	$-22x^2y + 22x^2 + 196xy^2 - 704xy + 394x - 98y^3 + 450y^2 - 500y + 108$
γ	$7x^4y^2 - 154x^4y + 371x^4 - 370y^2 + 1364y - 766$
$\{i, j\} = \{18, 26\}, k_{ij} = 2^5 \cdot 3^6$	
α	$44x^4 + 484y - 3448$
β	$-22x^7 + 11x^6y - 11x^6 - 77x^5y - 297x^5 + 33x^4y^2 + 198x^4y + 121x^4$ $+ 2098x^3 - 1049x^2y + 1049x^2 - 968xy^2 + 1975xy + 9271x + 484y^3 + 21y^2 - 6914y$ $- 2387$
γ	$-11x^4y^2 + 176x^4y + 583x^4 + 2985y^2 - 6048y - 17493$
$\{i, j\} = \{18, 27\}, k_{ij} = 2 \cdot 3 \cdot 5 \cdot 7$	
α	$-4x^4 + 48y - 13$
β	$8x^7 - 4x^4y^2 + 16x^4y - 4x^4 - 94x^3 - 60x + 24y^3 + 35y^2 - 242y + 17$
γ	$-12x^6 - 8x^5y + 20x^5 - 4x^4y - 8x^4 - 24x^2y^2 + 201x^2 + 24xy^2 + 94xy - 295x$ $- 24y^3 + 107y + 214$
$\{i, j\} = \{19, 25\}, k_{ij} = 2 \cdot 5$	
α	$319x^4 + 484y - 136$
β	$528x^7 + 11x^6y + 11x^6 - 99x^5y + 385x^5 + 330x^4y^2 - 1606x^4y + 451x^4$ $- 2273x^3 + 474x^2y + 474x^2 - 968xy^2 - 130xy + 2363x - 341y^3 - 707y^2 + 3673y - 126$
γ	$-1375x^6 - 550x^5y + 825x^5 - 11x^4y^2 + 2101x^4y - 1606x^4 - 825x^2y^2$ $- 275x^2y + 4550x^2 + 825xy^2 + 1600xy - 3225x + 825y^3 + 912y^2 - 6838y - 1975$
$\{i, j\} = \{19, 32\}, k_{ij} = 2^5 \cdot 3$	
α	$4x^4 - 2y + 14$
β	$-2x^7 + x^6y + x^6 - 8x^5y + 18x^5 + 5x^4y^2 - 19x^4y + 26x^4 - 18x^3$ $+ 9x^2y + 9x^2 + 4xy^2 - 36xy + 72x - 2y^3 + 33y^2 - 71y + 110$
γ	$-x^4y^2 + 14x^4y - 37x^4 - 17y^2 + 54y - 153$
$\{i, j\} = \{20, 34\}, k_{ij} = 2^2 \cdot 3^4$	
α	$-40x^4 + 200y + 364$
β	$-20x^7 - 10x^6y - 20x^6 - 70x^5y + 200x^5 - 30x^4y^2 - 20x^4y + 400x^4$ $+ 162x^3 + 81x^2y + 162x^2 + 400xy^2 - 153xy - 3580x + 200y^3 + 683y^2 - 2158y - 7140$
γ	$10x^4y^2 - 100x^4y + 440x^4 + 719y^2 - 630y - 7484$
$\{i, j\} = \{21, 32\}, k_{ij} = 2^2 \cdot 3$	
α	$14y - 12$
β	$7x^3y - 6x^3 + 14xy^2 - 12xy - 49x - 7y^3 + 6y^2 + 56y - 104$
γ	$-7x^3y + 6x^3 + 49x + 7y^3 - 6y^2 - 56y + 6$
$\{i, j\} = \{21, 33\}, k_{ij} = 5 \cdot 7$	
α	$64y + 48$
β	$32x^7 - 64x^6 - 32x^5 - 80x^4y - 280x^3 - 64x^2y^2 - 48x^2y + 512x^2 + 66x$ $- 64y^3 - 48y^2 + 620y + 511$
γ	$-32x^7 + 16x^4y + 280x^3 + 214x - 108y - 83$
$\{i, j\} = \{22, 31\}, k_{ij} = 2^3$	
α	$-y - 2$
β	$x^7 + x^6 + x^5 - 10x^3 + x^2y^2 + 2x^2y - 8x^2 - 6x - y^3 - 2y^2 - 3y - 2$
γ	$-x^7 - x^4y + 10x^3 - 4x + 11y + 6$
$\{i, j\} = \{23, 35\}, k_{ij} = 2^2 \cdot 3^2$	
α	$-2y + 8$
β	$x^7 + 2x^6 - 2x^5 + 3x^4y - 3x^4 - 11x^3 + 2x^2y^2 - 4x^2y - 26x^2 + 19x$ $+ 2y^3 - 6y^2 - 34y + 32$
γ	$-x^7 - x^4y + x^4 + 11x^3 + 3x + 12y - 12$
$\{i, j\} = \{23, 36\}, k_{ij} = 2^3 \cdot 3$	
α	$-2y + 6$
β	$x^7 + 2x^6 + 2x^4y - 2x^4 - 9x^3 + 2x^2y^2 - 2x^2y - 22x^2 + 4x + 2y^3 - 4y^2$ $- 21y + 33$
γ	$-x^7 + 9x^3 - 4x + y - 3$
$\{i, j\} = \{24, 33\}, k_{ij} = 2^3$	
α	$-2x^4 - 2y + 4$
β	$-x^7 + 4x^6 + 2x^5y - x^4y^2 - 3x^4y + 5x^4 + 4x^3 + 2x^2y^2 + 8x^2y$ $- 10x^2 - 8xy + x - 2y^3 - 2y^2 + 14y - 14$
γ	$x^7 + x^4y^2 - x^4y - x^4 - 4x^3 - x - 4y^2 + 4y + 2$
$\{i, j\} = \{24, 35\}, k_{ij} = 2^2 \cdot 5$	
α	$-2x^4 - 2y + 6$

β	$x^7 + 4x^6 - 2x^5y - x^4y^2 - 3x^4y + 3x^4 - 7x^3 + 2x^2y^2 + 12x^2y$
	$-18x^2 + 14xy - 4x - 2y^3 - 3y^2 + 18y - 23$
γ	$-x^7 + x^4y^2 - x^4y + x^4 + 7x^3 + 4x - 7y^2 + 12y - 3$
$\{i, j\} = \{25, 34\}, k_{ij} = 2^3 \cdot 5$	
α	$2x^3 - 6$
β	$-2x^5 + x^3y^2 - 7x^3 + 10x^2 + 6x - 3y^2 - 2y + 15$
γ	$2x^4 - x^3y^2 + 2x^3y + 5x^3 - 4x + 3y^2 - 8y - 17$
$\{i, j\} = \{25, 36\}, k_{ij} = 5^2 \cdot 7$	
α	$132x^4 + 2475y - 307$
β	$-33x^7 - 165x^6 + 66x^5y + 132x^5 + 66x^4y^2 + 330x^4y - 132x^4 + 3418x^3$
	$-495x^2y^2 - 3168x^2y + 3857x^2 + 495xy^2 - 3668xy - 6514x + 1485y^3 + 1678y^2$
	$-10492y - 5504$
γ	$33x^7 - 66x^4y^2 - 231x^4y - 3418x^3 + 6075x - 990y^3 + 995y^2 + 10303y - 132$
$\{i, j\} = \{26, 28\}, k_{ij} = 2^4 \cdot 3$	
α	$-2x^3 - 6$
β	$2x^5 + x^3y^2 - 5x^3 + 10x^2 - 8x + 3y^2 + 2y - 19$
γ	$2x^4 - x^3y^2 + 2x^3y + 9x^3 + 2x - 3y^2 + 8y + 23$
$\{i, j\} = \{26, 30\}, k_{ij} = 2^2 \cdot 3$	
α	$-2x^4 + 2y + 2$
β	$-x^7 - 2x^5y + x^4y^2 - x^4 - 4x^3 + 2x^2y^2 + 10x^2y - 12x^2 + 2xy^2 + 2xy$
	$-7x + 5y^2 + y$
γ	$x^7 - x^4y^2 - 2x^4y + x^4 + 4x^3 - 5x + 2y^3 + 9y^2 - y - 14$
$\{i, j\} = \{27, 29\}, k_{ij} = 2^5 \cdot 3^2 \cdot 5^2 \cdot 7$	
α	$-188x^4 - 3243y + 11831$
β	$47x^7 + 141x^6 + 94x^5y - 282x^5 + 94x^4y^2 - 376x^4y + 470x^4 + 4010x^3$
	$-1081x^2y^2 + 6768x^2y - 17157x^2 + 1081xy^2 + 1252xy - 14468x + 1081y^3$
	$+171y^2 - 32072y + 41827$
γ	$-47x^7 - 94x^4y^2 + 423x^4y - 329x^4 - 4010x^3 + 19595x - 2162y^3 + 6597y^2$
	$+6895y - 10202$
$\{i, j\} = \{27, 30\}, k_{ij} = 2^2 \cdot 3$	
α	2
β	$-2x^2 - y^2 - 2y + 7$
γ	$2x + y^2 - 7$
$\{i, j\} = \{27, 35\}, k_{ij} = 2^3 \cdot 3^3 \cdot 5$	
α	$y - 11$
β	$-x^7 - x^6 + 3x^5 - 5x^4y - x^4 + 29x^3 - x^2y^2 + 6x^2y + 40x^2 + xy^2 - 6xy$
	$+61x - 3y^3 + 19y^2 + 173y + 643$
γ	$x^7 + 4x^4y - x^4 - 29x^3 - 159x + 2y^3 - 13y^2 - 133y - 426$
$\{i, j\} = \{29, 32\}, k_{ij} = 2^2 \cdot 3 \cdot 5$	
α	$-2y + 12$
β	$x^7 + 2x^6 + 2x^5 - x^4y + 2x^4 - 16x^3 + 2x^2y^2 - 4x^2y - 40x^2 - 26x - 2y^3$
	$+8y^2 + 15y - 36$
γ	$-x^7 - x^4y + 2x^4 + 16x^3 - 6x + 17y - 32$
$\{i, j\} = \{29, 35\}, k_{ij} = 3^3 \cdot 5$	
α	$-208x^4 + 208y - 164$
β	$104x^7 + 416x^6 - 208x^5y + 104x^5 - 104x^4y^2 - 156x^4y + 624x^4 - 568x^3$
	$-208x^2y^2 + 468x^2y + 276x^2 + 1136xy - 1014x + 208y^3 - 316y^2 - 1102y + 369$
γ	$-104x^7 + 104x^4y^2 - 260x^4y + 208x^4 + 568x^3 + 446x - 568y^2 + 1762y - 709$
$\{i, j\} = \{31, 32\}, k_{ij} = 2^4$	
α	$y + 7$
β	$-x^7 - x^6 + 2x^5 + 4x^4y + 28x^3 - x^2y^2 - 4x^2y + 21x^2 + xy^2 + 4xy - 46x$
	$+3y^3 + 13y^2 - 86y + 23$
γ	$x^7 - 3x^4y - x^4 - 28x^3 - 3x - 2y^3 - 9y^2 + 65y + 8$
$\{i, j\} = \{33, 36\}, k_{ij} = 2 \cdot 5^2 \cdot 7$	
α	$64y + 624$
β	$-16x^7 - 64x^6 - 160x^5 + 8x^4y - 152x^4 + 616x^3 - 64x^2y^2 - 368x^2y$
	$+1920x^2 - 128xy^2 - 736xy + 3822x - 128y^2 - 1028y + 3045$
γ	$16x^7 - 72x^4y - 232x^4 - 616x^3 + 1250x - 64y^3 - 560y^2 + 1108y + 4019$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 10$ and for $x \leq -10$.

i	k_i	$\varphi_i(x)$	$x \geq 10$	$x \leq -10$
0	1	$x^6 - 3x^5 + 4x^4 - x^3 - 8x^2 + 9x - 1$	+	+
1	1	$x^4 + x - 1$	+	+
2	1	$x^4 - x^3 + 9x - 1$	+	+
3	1	$x^4 + x + 6$	+	+
4	1	$x^4 + x - 8$	+	+
5	1	$x^4 + x^3 - 7x - 1$	+	+
6	1	$x^4 - 8x^3 + 17x - 1$	+	+
7	1	$x^4 - x^3 - 3x^2 + 6x + 6$	+	+
8	1	$x^4 - x^3 + 3x^2 + 6x - 8$	+	+
9	1	$x^4 + x - 9$	+	+
10	1	$x^4 + x^3 - 3x^2 - 4x + 6$	+	+
11	1	$x^4 + x^3 + 3x^2 - 4x - 8$	+	+
12	1	$x^4 - 8x^3 - 12x^2 + 11x + 6$	+	+
13	1	$x^4 - 8x^3 + 12x^2 + 11x - 8$	+	+
14	1	$x^4 - x^3 - 6x^2 - 3x + 7$	+	+
15	1	$x^4 - x^3 + 6x^2 - 3x - 9$	+	+
16	1	$x^4 + x^3 - 6x^2 + 5x + 7$	+	+
17	1	$x^4 + x^3 + 6x^2 + 5x - 9$	+	+
18	1	$x^4 + 8x^3 - 12x^2 - 9x + 6$	+	+
19	1	$x^4 + 8x^3 + 12x^2 - 9x - 8$	+	+
20	1	$x^4 - 8x^3 + 24x^2 - 7x - 9$	+	+
21	-1	$x^6 - x^4 - 8x^2 - x + 1$	+	+
22	1	$x^6 + x^4 - 8x^2 + x - 1$	+	+
23	-1	$x^6 + 2x^4 - 5x^2 - x - 6$	+	+
24	1	$x^6 - 2x^4 - 5x^2 + x + 6$	+	+
25	1	$x^6 + 3x^5 + 4x^4 + x^3 - 8x^2 - 7x - 1$	+	+
26	-1	$x^6 - 3x^5 - x^4 + 5x^3 - 8x^2 + 4x + 8$	+	+
27	-1	$x^6 + 3x^5 + 5x^4 + 7x^3 - 2x^2 - 6x - 6$	+	+
28	1	$x^6 - 3x^5 + 7x^4 - 7x^3 - 2x^2 + 6x - 8$	+	+
29	1	$x^6 - 5x^4 + 4x^2 + x + 7$	+	+
30	1	$x^6 + 3x^5 + x^4 - 5x^3 - 8x^2 - 4x + 6$	+	+
31	1	$x^6 + 3x^5 + 7x^4 + 7x^3 - 2x^2 - 4x - 8$	+	+
32	1	$x^6 + 6x^5 + 13x^4 + 8x^3 - 8x^2 - 15x - 1$	+	+
33	-1	$x^6 - 6x^5 + 8x^4 + 4x^3 - 17x^2 + 9x + 8$	+	+
34	-1	$x^6 + 3x^5 - 4x^4 - 11x^3 - 2x^2 + 3x + 9$	+	+
35	-1	$x^6 + 6x^5 + 8x^4 - 4x^3 - 17x^2 - 11x + 8$	+	+
36	-1	$x^6 + 6x^5 + 14x^4 + 20x^3 + 7x^2 - 11x - 6$	+	+

Table 4: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 4, grouped by p, k and the value of B_p .

p	k	B_p	classes (a, b)
2	5	1	(3, 19)
2	6	1	(0, 39), (1, 25), (2, 9), (4, 3), (5, 37), (7, 31), (10, 33), (11, 43), (12, 27), (13, 29), (15, 23), (16, 23), (17, 9), (18, 57), (19, 35), (20, 51), (21, 21), (23, 15), (26, 17), (27, 27), (28, 11), (29, 13), (31, 7), (32, 7), (33, 57), (34, 41), (36, 35), (37, 5), (39, 63), (42, 1), (43, 11), (44, 59), (45, 61), (47, 55), (48, 55), (49, 41), (50, 25), (51, 3), (52, 19), (53, 53), (55, 47), (58, 49), (59, 59), (60, 43), (61, 45), (63, 39)
2	7	1	(8, 127), (9, 17), (14, 93), (24, 47), (25, 1), (30, 13), (40, 95), (41, 113), (46, 61), (54, 85), (56, 15), (57, 97), (62, 109), (72, 63), (73, 81), (78, 29), (88, 111), (89, 65), (94, 77), (102, 101), (104, 31), (105, 49), (110, 125), (120, 79), (121, 33), (126, 45)
2	8	1	(6, 69), (22, 245), (38, 165), (70, 5), (86, 181), (118, 21), (134, 197), (150, 117), (166, 37), (198, 133), (214, 53), (246, 149)
3	3	-1	(0, 16), (1, 11), (4, 8), (7, 5), (8, 16), (9, 25), (10, 2), (13, 26), (17, 16), (18, 7), (22, 17), (25, 14), (26, 16)

p	k	B_p	classes (a, b)
3	4	-1	(2, 61), (3, 64), (5, 52), (6, 58), (11, 34), (12, 46), (14, 25), (15, 40), (16, 50), (19, 47), (20, 7), (21, 28), (29, 61), (30, 10), (32, 52), (33, 4), (38, 34), (39, 73), (41, 25), (42, 67), (43, 23), (46, 20), (47, 7), (48, 55), (51, 49), (56, 61), (57, 37), (59, 52), (60, 31), (65, 34), (66, 19), (68, 25), (69, 13), (70, 77), (73, 74), (74, 7), (75, 1)
3	5	-1	(23, 160), (24, 103), (50, 79), (77, 241), (78, 238), (104, 160), (105, 184), (131, 79), (185, 160), (186, 22), (212, 79), (240, 157)
3	6	-1	(158, 484), (159, 76), (239, 241), (401, 484), (402, 319), (482, 241), (644, 484), (645, 562), (725, 241)
5	2	1	(2, 2), (3, 23), (7, 4), (7, 9), (7, 12), (7, 14), (7, 24), (8, 3), (12, 22), (18, 1), (18, 6), (18, 11), (18, 13), (18, 16), (18, 21), (22, 17), (23, 18)
5	3	1	(13, 8), (17, 7), (38, 33), (42, 57), (63, 58), (67, 107), (88, 83), (92, 32), (107, 19), (107, 44), (107, 69), (107, 94), (107, 119), (113, 108), (117, 82)
7	1	1	(0, 2), (1, 5), (4, 6), (5, 2), (6, 2)
7	2	1	(3, 37), (4, 35), (4, 36), (9, 3), (10, 44), (11, 0), (11, 29), (16, 10), (17, 2), (18, 14), (18, 22), (23, 17), (24, 9), (25, 15), (25, 28), (30, 24), (31, 16), (32, 8), (32, 42), (37, 31), (38, 23), (39, 1), (39, 7), (44, 38), (45, 30), (46, 21), (46, 43)
7	3	1	(2, 143), (51, 192), (100, 241), (149, 290), (198, 339), (247, 45), (296, 94)